

Honours Multivariate Analysis

Lecture 5 - Multivariate Inference

Stefan S. Britz

Department of Statistical Sciences
University of Cape Town



Course Outline

- ① Introduction, Examples of Multivariate Data †
- ② TOOLS
 - ① Visualization and Summary Statistics †
 - ② Singular Value Decomposition, Eigenvalue Decomposition and Spectral Decomposition revisited †
 - ③ The Multivariate Normal Distribution †
 - ④ Multivariate Maximum Likelihood Estimation †
 - ⑤ **Multivariate Inference** †
- ③ EXPLORATORY ANALYSIS
 - ① Principal Component Analysis §
 - ② Factor Analysis §
 - ③ Correspondence Analysis §
- ④ CONFIRMATORY ANALYSIS
 - ① For grouped Multivariate Data:
 - ① Manova †
 - ② Discriminant Analysis §
 - ② Regression
 - ① Multivariate Regression §
 - ② Canonical Correlation Analysis §

†Mr Stefan Britz

§Mr Miguel Rodo

Inference about a Mean Vector (Univariate intro)

Goal: Test whether a given vector of values is a plausible mean of a multivariate normal distribution, given a set of observations. First, a reminder of the univariate case:

Inference about a Mean Vector (Univariate intro)

Goal: Test whether a given vector of values is a plausible mean of a multivariate normal distribution, given a set of observations. First, a reminder of the univariate case:

Suppose we have a random sample $X_1, X_2, \dots, X_n$ from a normal population with unknown mean μ and unknown variance σ^2 . We can test the univariate hypothesis, $H_0: \mu = \mu_0$ vs. $H_1: \mu \neq \mu_0$ using a t-statistic:

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim t_{n-1}$$

where $\bar{X} = \frac{1}{n} \sum_{j=1}^n X_j$ and $s^2 = \frac{1}{n-1} \sum_{j=1}^n (X_j - \bar{X})^2$.

Inference about a Mean Vector (Univariate intro)

Goal: Test whether a given vector of values is a plausible mean of a multivariate normal distribution, given a set of observations. First, a reminder of the univariate case:

Suppose we have a random sample $X_1, X_2, \dots, X_n$ from a normal population with unknown mean μ and unknown variance σ^2 . We can test the univariate hypothesis, $H_0: \mu = \mu_0$ vs. $H_1: \mu \neq \mu_0$ using a t-statistic:

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim t_{n-1}$$

where $\bar{X} = \frac{1}{n} \sum_{j=1}^n X_j$ and $s^2 = \frac{1}{n-1} \sum_{j=1}^n (X_j - \bar{X})^2$.

We will reject H_0 when $|t|$ is large, which is equivalent to rejecting H_0 when

$$t^2 = \frac{(\bar{X} - \mu_0)^2}{s^2/n} = n(\bar{X} - \mu_0)(s^2)^{-1}(\bar{X} - \mu_0)$$

is large.

Univariate Inference about the Mean

For an observed $\bar{X}$ and s^2 , reject H_0 in favour of H_1 if

$$n(\bar{X} - \mu_0)(s^2)^{-1}(\bar{X} - \mu_0) > t_{n-1}^2(\alpha/2)$$

where $t_{n-1}(\alpha/2)$ denotes the upper $100(\alpha/2)^{th}$ percentile of t_{n-1} .

Univariate Inference about the Mean

For an observed $\bar{X}$ and s^2 , reject H_0 in favour of H_1 if

$$n(\bar{X} - \mu_0)(s^2)^{-1}(\bar{X} - \mu_0) > t_{n-1}^2(\alpha/2)$$

where $t_{n-1}(\alpha/2)$ denotes the upper $100(\alpha/2)^{th}$ percentile of t_{n-1} .

Note that there is actually a whole range of plausible values in support of $H_0: \mu = \mu_0$ since

$$\{\text{Do not reject } H_0: \mu = \mu_0 \text{ at level } \alpha\} \text{ or } \left| \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \right| \leq t_{n-1}(\alpha/2)$$

is equivalent to

$$\{\mu_0 \text{ lies in the } 100(1 - \alpha)\% \text{ confidence interval } \bar{x} \pm t_{n-1}(\alpha/2) \frac{s}{\sqrt{n}}\} \text{ or}$$

$$\bar{x} - t_{n-1}(\alpha/2) \frac{s}{\sqrt{n}} \leq \mu_0 \leq \bar{x} + t_{n-1}(\alpha/2) \frac{s}{\sqrt{n}}$$

Univariate Inference about the Mean

- Since $\bar{X}$ and s are random variables, the confidence interval is a random interval.
- The probability that the interval contains μ is $1 - \alpha$.
- This is the same as saying that among large numbers of such independent intervals, approximately $100(1 - \alpha)\%$ of them will contain μ .

Hotelling's T^2 statistic

Now consider $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ drawn from $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$

The multivariate analogue of the t^2 statistic to test $H_0: \boldsymbol{\mu} = \boldsymbol{\mu}_0$ vs. $H_1: \boldsymbol{\mu} \neq \boldsymbol{\mu}_0$ is Hotelling's T^2 :

$$T^2 = n(\bar{\mathbf{X}} - \boldsymbol{\mu}_0)' \mathbf{S}^{-1} (\bar{\mathbf{X}} - \boldsymbol{\mu}_0)$$

$$\text{where } \mathbf{X}_{p \times 1} = \frac{1}{n} \sum_{j=1}^n \mathbf{X}_j,$$

$$\mathbf{S}_{p \times p} = \frac{1}{n-1} \sum_{j=1}^n (\mathbf{X}_j - \bar{\mathbf{X}})(\mathbf{X}_j - \bar{\mathbf{X}})', \text{ and}$$

$$\boldsymbol{\mu}_{0:p \times 1} = \begin{bmatrix} \mu_{10} \\ \mu_{20} \\ \vdots \\ \mu_{p0} \end{bmatrix}$$

Hotelling's T^2 statistic

It can be shown that this statistic is proportional to the F -distribution with p and $n - p$ degrees of freedom, such that

$$T^2 \sim \frac{(n-1)p}{n-p} F_{p, n-p}$$

$\implies \Pr \left[T^2 > \frac{(n-1)p}{n-p} F_{p, n-p}(\alpha) \right] = \alpha$ where $F_{p, n-p}(\alpha)$ refers to the upper $(100\alpha)\%$ of the $F_{p, n-p}$ distribution.

Hotelling's T^2 statistic

It can be shown that this statistic is proportional to the F -distribution with p and $n - p$ degrees of freedom, such that

$$T^2 \sim \frac{(n-1)p}{n-p} F_{p, n-p}$$

$\implies \Pr \left[T^2 > \frac{(n-1)p}{n-p} F_{p, n-p}(\alpha) \right] = \alpha$ where $F_{p, n-p}(\alpha)$ refers to the upper $(100\alpha)\%$ of the $F_{p, n-p}$ distribution.

Therefore, we reject H_0 if the observed T^2 is greater than $\frac{(n-1)p}{n-p} F_{p, n-p}(\alpha)$.

Hotelling's T^2 statistic

It can be shown that this statistic is proportional to the F -distribution with p and $n - p$ degrees of freedom, such that

$$T^2 \sim \frac{(n-1)p}{n-p} F_{p,n-p}$$

$\implies \Pr \left[T^2 > \frac{(n-1)p}{n-p} F_{p,n-p}(\alpha) \right] = \alpha$ where $F_{p,n-p}(\alpha)$ refers to the upper $(100\alpha)\%$ of the $F_{p,n-p}$ distribution.

Therefore, we reject H_0 if the observed T^2 is greater than $\frac{(n-1)p}{n-p} F_{p,n-p}(\alpha)$.

We could rather calculate the associated p -value to measure weight of evidence against H_0 . Much more informative!

$p = \Pr(T^2 \geq T^*)$ for some observed test statistic T^*

Hotelling's T^2 statistic

Note that we can write T^2 as

$$\begin{aligned} T^2 &= \sqrt{n}(\bar{\mathbf{X}} - \boldsymbol{\mu}_0)' \left(\frac{\sum_{j=1}^n (\mathbf{X}_j - \bar{\mathbf{X}})(\mathbf{X}_j - \bar{\mathbf{X}})'}{n-1} \right)^{-1} \sqrt{n}(\bar{\mathbf{X}} - \boldsymbol{\mu}_0) \\ &= (\text{MVN random vector})' \left(\frac{\text{Wishart random matrix}}{df} \right)^{-1} (\text{MVN random vector}) \\ &= N_p(\mathbf{0}, \boldsymbol{\Sigma})' \left[\frac{1}{n-1} W_{p,n-1}(\boldsymbol{\Sigma}) \right]^{-1} N_p(\mathbf{0}, \boldsymbol{\Sigma}) \end{aligned}$$

- N_p and $W_{p,n-1}$ are independent $\implies$ the product will yield the joint density.
- From this we can derive the F -distribution of T^2 .

Example

Example 5.2 on page 214 of Johnson & Wichern. Solution in `Lecture5.R`.

Perspiration from 20 similar participants were analysed, with three components measured: X_1 = sweat rate, X_2 = sodium content, and X_3 = potassium content.

```
sweat <- read.table('T5-1.dat')
head(sweat, 4)
```

	V1	V2	V3
1	3.7	48.5	9.3
2	5.7	65.1	8.0
3	3.8	47.2	10.9
4	3.2	53.2	12.0

```
n <- nrow(sweat); p <- ncol(sweat)
(xbar <- matrix(apply(sweat, 2, mean), 3))
(S <- cov(sweat))
```

		[,1]	
[1,]		4.640	
[2,]		45.400	
[3,]		9.965	
	V1	V2	V3
V1	2.879368	10.0100	-1.809053
V2	10.010000	199.7884	-5.640000
V3	-1.809053	-5.6400	3.627658

Example

Suppose we want to test the hypothesis $H_0: \boldsymbol{\mu} = \begin{bmatrix} 4 \\ 50 \\ 10 \end{bmatrix}$ vs $H_1: \boldsymbol{\mu} \neq \begin{bmatrix} 4 \\ 50 \\ 10 \end{bmatrix}$. Using this vector of values for $\boldsymbol{\mu}_0$, we can calculate T^2 and its associated p-value.

```
(mu0 <- matrix(c(4, 50, 10), 3))
```

```
      [,1]  
[1,]    4  
[2,]   50  
[3,]   10
```

```
# Calculate Hotelling's T^2  
(T2 <- n*t(xbar - mu0)%*%solve(S)%*%(xbar - mu0))
```

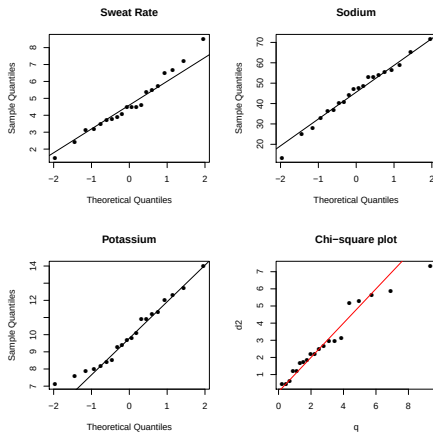
```
      [,1]  
[1,] 9.738773
```

```
# Calculate p-value  
1 - pf(T2*(n-p)/((n-1)*p), p, n-p)
```

```
      [,1]  
[1,] 0.0649
```

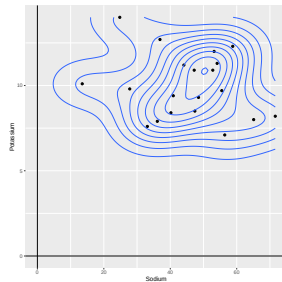
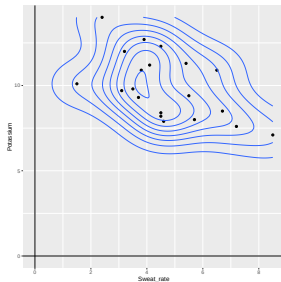
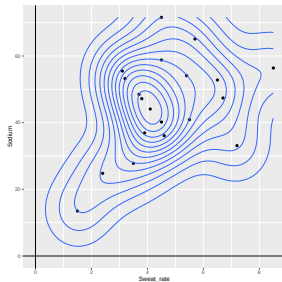
Example

Checking multivariate normality for this example:



Example

Checking multivariate normality for this example:



Invariance of the T^2 statistic

Let $\mathbf{Y} = \mathbf{C}\mathbf{X} + \mathbf{d}$, where $\mathbf{C}_{p \times p}$ is non-singular and $\mathbf{d}_{p \times 1}$ is a vector of constants.

Testing

$$H_0: \boldsymbol{\mu}_Y = \mathbf{C}\boldsymbol{\mu}_0 + \mathbf{d} \text{ vs } H_1: \boldsymbol{\mu}_Y \neq \mathbf{C}\boldsymbol{\mu}_0 + \mathbf{d}$$

is equivalent to testing

$$H_0: \boldsymbol{\mu}_X = \boldsymbol{\mu}_0 \text{ vs } H_1: \boldsymbol{\mu}_X \neq \boldsymbol{\mu}_0$$

To show this, we first note that if $\mathbf{X} \sim N_p(\boldsymbol{\mu}_X, \boldsymbol{\Sigma})$, then

$\mathbf{Y} \sim N_p(\mathbf{C}\boldsymbol{\mu}_X + \mathbf{d}, \mathbf{C}\boldsymbol{\Sigma}\mathbf{C}')$ such that $\bar{\mathbf{Y}} = \mathbf{C}\bar{\mathbf{X}} + \mathbf{d}$ and $\mathbf{S}_Y = \mathbf{C}\mathbf{S}_X\mathbf{C}'$.

Invariance of the T^2 statistic

Now

$$\begin{aligned}T_Y^2 &= n(\bar{Y} - C\mu_0 - d)'S_Y^{-1}(\bar{Y} - C\mu_0 - d) \\&= n(C\bar{X} + d - C\mu_0 - d)'(CS_X C')^{-1}(C\bar{X} + d - C\mu_0 - d) \\&= n[C(\bar{X} - \mu_0)]'(C')^{-1}S_X^{-1}C^{-1}[C(\bar{X} - \mu_0)] \\&= n(\bar{X} - \mu_0)'C'(C')^{-1}S_X^{-1}C^{-1}C(\bar{X} - \mu_0) \\&= n(\bar{X} - \mu_0)'S_X^{-1}(\bar{X} - \mu_0) \\&= T_X^2\end{aligned}$$

This shows that Hotelling's T^2 statistic is invariant under changes in the unit of measurement for X of the form $Y = CX + d$.

Hotelling's T^2 statistic and Likelihood Ratio tests

- A general frequentist approach to inference is the likelihood ratio test.
- It compares the maximum of the likelihood when the parameter space is restricted under H_0 to the maximum of the likelihood when parameters are unrestricted.
- A likelihood ratio test of $H_0: \theta \in \Theta_0$ rejects H_0 in favour of $H_1: \theta \notin \Theta_0$ if

$$\Lambda = \frac{\max_{\theta \in \Theta_0} L(\theta)}{\max_{\theta \in \Theta} L(\theta)} < c$$

for a suitably chosen constant c .

- It can be shown that for large sample sizes n , the value $-2 \log(\Lambda)$ is approximately distributed as $\chi^2_{\nu - \nu_0}$, where $\nu = \dim(\Theta)$ and $\nu_0 = \dim(\Theta_0)$.

Hotelling's T^2 statistic and Likelihood Ratio tests

Consider the hypothesis $H_0: \boldsymbol{\mu} = \boldsymbol{\mu}_0$ against the alternative $H_1: \boldsymbol{\mu} \neq \boldsymbol{\mu}_0$ for $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

Now the unrestricted parameter space is

$$\Theta = \{-\infty < \mu_1 < \infty, \dots, -\infty < \mu_p < \infty, \sigma_{11}, \sigma_{12}, \dots, \sigma_{1p}, \sigma_{22}, \dots, \sigma_{pp}: \boldsymbol{\Sigma} > 0\}$$

$$\nu = \dim(\Theta) = p + \frac{p(p+1)}{2}$$

and the parameter space under $H_0: \boldsymbol{\mu} = \boldsymbol{\mu}_0$ is

$$\Theta = \{\mu_1 = \mu_{10}, \dots, \mu_p = \mu_{p0}, \quad \sigma_{11}, \sigma_{12}, \dots, \sigma_{1p}, \sigma_{22}, \dots, \sigma_{2p}, \dots, \sigma_{pp}: \boldsymbol{\Sigma} > 0\}$$

$$\nu_0 = \dim(\Theta_0) = \frac{p(p+1)}{2}$$

Hotelling's T^2 statistic and Likelihood Ratio tests

Last week we saw that the maximum of the multivariate normal likelihood as μ and Σ are varied over their possible values is given by

$$\max_{\theta \in \Theta} L(\mu, \Sigma) = (2\pi)^{-\frac{np}{2}} \left| \frac{1}{n} \mathbf{A} \right|^{-\frac{n}{2}} e^{-\frac{np}{2}}$$

Under H_0 , we wish to find the most likely value of Σ that with $\mu = \mu_0$ fixed, would have led to the observed data. This is equivalent to maximizing

$$L(\mu_0, \Sigma) = \frac{1}{(2\pi)^{\frac{np}{2}} |\Sigma|^{\frac{n}{2}}} \exp \left[-\frac{1}{2} \sum_{j=1}^n (\mathbf{x}_j - \mu_0)' \Sigma^{-1} (\mathbf{x}_j - \mu_0) \right]$$

Hotelling's T^2 statistic and Likelihood Ratio tests

After taking logs we have

$$\begin{aligned}l(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}) &= -\frac{np}{2} \log(2\pi) - \frac{n}{2} \log |\boldsymbol{\Sigma}| - \frac{1}{2} \sum_{j=1}^n (\mathbf{x}_j - \boldsymbol{\mu}_0)' \boldsymbol{\Sigma}^{-1} (\mathbf{x}_j - \boldsymbol{\mu}_0) \\&= -\frac{np}{2} \log(2\pi) + \frac{n}{2} \log |\boldsymbol{\Sigma}^{-1}| - \frac{1}{2} \sum_{j=1}^n \text{tr} [(\mathbf{x}_j - \boldsymbol{\mu}_0)(\mathbf{x}_j - \boldsymbol{\mu}_0)' \boldsymbol{\Sigma}^{-1}]\end{aligned}$$

Differentiating and setting equal to zero:

$$\frac{\partial l}{\partial \boldsymbol{\Sigma}^{-1}} = \frac{n}{2} \boldsymbol{\Sigma} - \frac{1}{2} \sum_{j=1}^n (\mathbf{x}_j - \boldsymbol{\mu}_0)(\mathbf{x}_j - \boldsymbol{\mu}_0)' = 0$$

$$\begin{aligned}\hat{\boldsymbol{\Sigma}}_0 &= \frac{1}{n} \sum_{j=1}^n (\mathbf{x}_j - \boldsymbol{\mu}_0)(\mathbf{x}_j - \boldsymbol{\mu}_0)' \\&= \frac{1}{n} \mathbf{A} + (\bar{\mathbf{x}} - \boldsymbol{\mu}_0)(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)'\end{aligned}$$

It can again be shown that this indeed yields a maximum for $L(\boldsymbol{\mu}_0, \boldsymbol{\Sigma})$

Hotelling's T^2 statistic and Likelihood Ratio tests

Now we can write

$$\begin{aligned}\sum_{j=1}^n (\mathbf{x}_j - \boldsymbol{\mu}_0)' \hat{\boldsymbol{\Sigma}}_0^{-1} (\mathbf{x}_j - \boldsymbol{\mu}_0) &= \sum_{j=1}^n \text{tr} \left[(\mathbf{x}_j - \boldsymbol{\mu}_0)(\mathbf{x}_j - \boldsymbol{\mu}_0)' \hat{\boldsymbol{\Sigma}}_0^{-1} \right] \\ &= \text{tr} \left[\sum_{j=1}^n (\mathbf{x}_j - \boldsymbol{\mu}_0)(\mathbf{x}_j - \boldsymbol{\mu}_0)' \hat{\boldsymbol{\Sigma}}_0^{-1} \right] \\ &= \text{tr} \left[n \sum_{j=1}^n \left[\frac{1}{n} (\mathbf{x}_j - \boldsymbol{\mu}_0)(\mathbf{x}_j - \boldsymbol{\mu}_0)' \right] \hat{\boldsymbol{\Sigma}}_0^{-1} \right] \\ &= \text{tr} \left[n \hat{\boldsymbol{\Sigma}}_0 \hat{\boldsymbol{\Sigma}}_0^{-1} \right] = \text{tr}(n \mathbf{I}_p) = np\end{aligned}$$

so that

$$\max_{\boldsymbol{\theta} \in \Theta_0} L(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = (2\pi)^{-\frac{np}{2}} \left| \hat{\boldsymbol{\Sigma}}_0 \right|^{-\frac{n}{2}} e^{-\frac{np}{2}}$$

Hotelling's T^2 statistic and Likelihood Ratio tests

Therefore, the likelihood ratio statistic for $H_0: \boldsymbol{\mu} = \boldsymbol{\mu}_0$ is

$$\begin{aligned}\Lambda &= \frac{\max_{\theta \in \Theta_0} L(\theta)}{\max_{\theta \in \Theta} L(\theta)} \\&= \frac{(2\pi)^{-\frac{np}{2}} \left| \hat{\boldsymbol{\Sigma}}_0 \right|^{-\frac{n}{2}} e^{-\frac{np}{2}}}{(2\pi)^{-\frac{np}{2}} \left| \frac{1}{n} \mathbf{A} \right|^{-\frac{n}{2}} e^{-\frac{np}{2}}} \\&= \left(\frac{\left| \frac{1}{n} \mathbf{A} \right|}{\left| \hat{\boldsymbol{\Sigma}}_0 \right|} \right)^{\frac{n}{2}} \\&= \left(\frac{\left| \hat{\boldsymbol{\Sigma}} \right|}{\left| \hat{\boldsymbol{\Sigma}}_0 \right|} \right)^{\frac{n}{2}}\end{aligned}$$

Wilk's Lambda

If the observed value of this likelihood ratio is too small, $H_0: \boldsymbol{\mu} = \boldsymbol{\mu}_0$ is unlikely to be true and so we will reject H_0 if

$$\Lambda = \left(\frac{|\hat{\boldsymbol{\Sigma}}|}{|\hat{\boldsymbol{\Sigma}}_0|} \right)^{\frac{n}{2}} = \left(\frac{|\mathbf{A}|}{|\mathbf{A} + n(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)'|} \right)^{\frac{n}{2}} < c_\alpha$$

where c_α is the lower $(100\alpha)^{\text{th}}$ percentile of the distribution of Λ .

We now define Wilk's lambda as

$$\Lambda^{\frac{2}{n}} = \frac{|\hat{\boldsymbol{\Sigma}}|}{|\hat{\boldsymbol{\Sigma}}_0|}$$

The link between Wilk's lambda and Hotelling's T^2

If $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ is a random sample from an $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ population, then the test based on T^2 is equivalent to the likelihood ratio test of $H_0: \boldsymbol{\mu} = \boldsymbol{\mu}_0$ because

$$\Lambda^{\frac{2}{n}} = \left(1 + \frac{T^2}{n-1}\right)^{-1}$$

To prove this, we start by constructing a matrix

$$\mathbf{B}_{(p+1) \times (p+1)} = \begin{bmatrix} \mathbf{A} & \sqrt{n}(\bar{\mathbf{x}} - \boldsymbol{\mu}_0) \\ \sqrt{n}(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)' & -1 \end{bmatrix}$$

and then apply Theorem A.4 in Appendix A (see notes for derivation).

The link between Wilk's lambda and Hotelling's T^2

$$\Lambda^{\frac{2}{n}} = \left(1 + \frac{1}{n-1}T^2\right)^{-1}$$

What this shows is that T^2 can be computed from two determinants,

$$\begin{aligned}T^2 &= \frac{(n-1)|\hat{\Sigma}_0|}{|\hat{\Sigma}|} - (n-1) \\&= \frac{(n-1)|\sum_{j=1}^n (\mathbf{x}_j - \boldsymbol{\mu}_0)(\mathbf{x}_j - \boldsymbol{\mu}_0)'|}{|\sum_{j=1}^n (\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})'|} - (n-1)\end{aligned}$$

which does not depend on the inverse of $\mathbf{S}$.

Multivariate Confidence Regions

Inference about a parameter (set) goes hand-in-hand with confidence intervals (regions) around the parameter (set)

- Univariate: Confidence **interval**. $\theta \pm c$
- Multivariate: Confidence **region**

The region, $R(\mathbf{X})$, is determined by the data, where $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n]'$ is the data matrix.

$R(\mathbf{X})$ is said to be the $100(1 - \alpha)\%$ “confidence region” if, before the sample is selected, the probability that $R(\mathbf{X})$ will cover the true $\boldsymbol{\theta}$ is $1 - \alpha$.

Multivariate Confidence Regions

For the mean, $\boldsymbol{\mu}$,

$$\Pr \left[n(\bar{\mathbf{X}} - \boldsymbol{\mu})' \mathbf{S}^{-1} (\bar{\mathbf{X}} - \boldsymbol{\mu}) \leq \frac{(n-1)p}{n-p} F_{p, n-p}(\alpha) \right] = 1 - \alpha$$

which implies that $\bar{\mathbf{X}}$ will be within $\left[\frac{(n-1)p}{n-p} F_{p, n-p}(\alpha) \right]^{\frac{1}{2}}$ of $\boldsymbol{\mu}$ with probability $1 - \alpha$, if distance is defined in terms of $n\mathbf{S}^{-1}$.

Multivariate Confidence Regions

For the mean, $\boldsymbol{\mu}$,

$$\Pr \left[n(\bar{\mathbf{X}} - \boldsymbol{\mu})' \mathbf{S}^{-1} (\bar{\mathbf{X}} - \boldsymbol{\mu}) \leq \frac{(n-1)p}{n-p} F_{p, n-p}(\alpha) \right] = 1 - \alpha$$

which implies that $\bar{\mathbf{X}}$ will be within $\left[\frac{(n-1)p}{n-p} F_{p, n-p}(\alpha) \right]^{\frac{1}{2}}$ of $\boldsymbol{\mu}$ with probability $1 - \alpha$, if distance is defined in terms of $n\mathbf{S}^{-1}$.

For a particular sample, we can compute $\bar{\mathbf{x}}$ and $\mathbf{S}$ and then

$$n(\bar{\mathbf{x}} - \boldsymbol{\mu})' \mathbf{S}^{-1} (\bar{\mathbf{x}} - \boldsymbol{\mu}) \leq \frac{(n-1)p}{n-p} F_{p, n-p}(\alpha)$$

will define the region $R(\mathbf{X})$. This region will be a p -dimensional ellipsoid centred at $\bar{\mathbf{x}}$ and will present the $100(1 - \alpha)\%$ confidence region for $\boldsymbol{\mu}$.

Multivariate Confidence Regions

When $p \geq 4$ we cannot graph the joint confidence region for $\boldsymbol{\mu}$...

But we can calculate the axes of the confidence ellipsoid and their relative lengths similarly to how we did it previously – by finding the eigenvalues (λ_i) and the eigenvectors ($\mathbf{e}_i$) of $\mathbf{S}$!

The axes directions of the ellipsoid

$$n(\bar{\mathbf{X}} - \boldsymbol{\mu})' \mathbf{S}^{-1} (\bar{\mathbf{X}} - \boldsymbol{\mu}) \leq c^2 = \frac{(n-1)p}{n-p} F_{p, n-p}(\alpha)$$

are given by $\mathbf{e}_i$,

and the half-length by $\frac{\sqrt{\lambda_i} c_i}{\sqrt{n}} = \sqrt{\lambda_i} \sqrt{\frac{(n-1)p}{n(n-p)} F_{p, n-p}(\alpha)}$

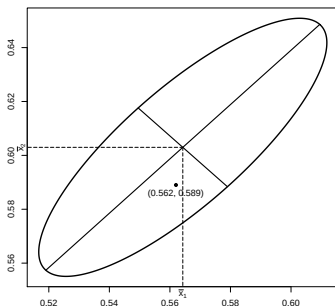
Multivariate Confidence Regions

Beginning at the centre $\bar{x}$, the axes of the confidence ellipsoids are

$$\pm \sqrt{\lambda_i} \sqrt{\frac{p(n-1)}{n(n-p)} F_{p,n-p}(\alpha)} \mathbf{e}_i \text{ where } \mathbf{S} \mathbf{e}_i = \lambda_i \mathbf{e}_i, i = 1, \dots, p$$

Thus the ratios of the λ_i will help us identify the relative amounts of elongation along pairs of axes.

Example 5.3 on page 221 of Johnson & Wichern. Solution in [Lecture5.R](#).



Simultaneous Confidence Intervals

If $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ is a random sample from $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ then, simultaneously for all $\mathbf{a}$, the interval

$$\left(\mathbf{a}' \bar{\mathbf{X}} - \sqrt{\frac{p(n-1)}{n(n-p)} F_{p, n-p}(\alpha)} \mathbf{a}' \mathbf{S} \mathbf{a} ; \mathbf{a}' \bar{\mathbf{X}} + \sqrt{\frac{p(n-1)}{n(n-p)} F_{p, n-p}(\alpha)} \mathbf{a}' \mathbf{S} \mathbf{a} \right)$$

will contain $\mathbf{a}' \boldsymbol{\mu}$ with probability $1 - \alpha$.

These are referred to as T^2 intervals since the coverage probability is determined by the distribution of T^2 .

Simultaneous Confidence Intervals

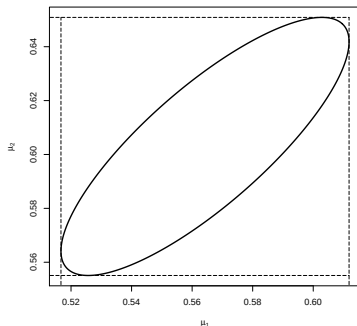
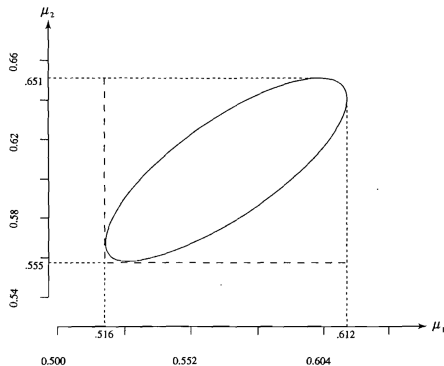
By choosing $\mathbf{a}' = [1, 0, \dots, 0]$, $\mathbf{a}' = [0, 1, \dots, 0]$, and so on, we obtain p confidence intervals of the form

$$\begin{aligned}\bar{x}_1 - \sqrt{\frac{p(n-1)}{(n-p)} F_{p,n-p}(\alpha)} \sqrt{\frac{s_{11}}{n}} &\leq \mu_1 \leq \bar{x}_1 + \sqrt{\frac{p(n-1)}{(n-p)} F_{p,n-p}(\alpha)} \sqrt{\frac{s_{11}}{n}} \\ \bar{x}_2 - \sqrt{\frac{p(n-1)}{(n-p)} F_{p,n-p}(\alpha)} \sqrt{\frac{s_{22}}{n}} &\leq \mu_2 \leq \bar{x}_2 + \sqrt{\frac{p(n-1)}{(n-p)} F_{p,n-p}(\alpha)} \sqrt{\frac{s_{22}}{n}} \\ &\vdots \\ \bar{x}_p - \sqrt{\frac{p(n-1)}{(n-p)} F_{p,n-p}(\alpha)} \sqrt{\frac{s_{pp}}{n}} &\leq \mu_p \leq \bar{x}_p + \sqrt{\frac{p(n-1)}{(n-p)} F_{p,n-p}(\alpha)} \sqrt{\frac{s_{pp}}{n}}\end{aligned}$$

that all hold simultaneously with probability $1 - \alpha$.

Simultaneous Confidence Intervals

Example 5.4 (continued)



Simultaneous versus one-at-a-time intervals

One-at-a-time intervals, ignoring the covariance structure of the p variables:

$$\bar{x}_i - t_{n-1}(\alpha/2)\sqrt{\frac{s_{ii}}{n}} \leq \mu_i \leq \bar{x}_i + t_{n-1}(\alpha/2)\sqrt{\frac{s_{ii}}{n}}$$

as opposed to the simultaneous interval

$$\bar{x}_i - \sqrt{\frac{p(n-1)}{(n-p)}F_{p,n-p}(\alpha)}\sqrt{\frac{s_{ii}}{n}} \leq \mu_i \leq \bar{x}_i + \sqrt{\frac{p(n-1)}{(n-p)}F_{p,n-p}(\alpha)}\sqrt{\frac{s_{ii}}{n}}.$$

Prior to sampling, the i^{th} individual interval has probability $1 - \alpha$ of covering μ_i but we cannot say that ALL intervals will contain their respective μ_i with probability $1 - \alpha$. Why not?

Simultaneous versus one-at-a-time intervals

Consider the special case where $\Sigma = \text{diag}(\sigma_{11}, \sigma_{22}, \dots, \sigma_{pp})$, which implies that the p variables are independent. Thus

$$\Pr(\text{all } t\text{-intervals contain the } \mu'_i s) = (1-\alpha)(1-\alpha) \dots (1-\alpha) = (1-\alpha)^p < (1-\alpha)$$

$$\forall \alpha > 0, p \geq 2$$

Therefore we need wider intervals and hence the different multipliers of $\sqrt{\frac{s_{ii}}{n}}$.

For example, for $1 - \alpha = 0.95, n = 15, p = 4$,

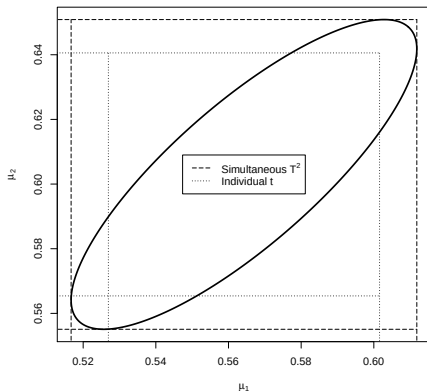
$$\sqrt{\frac{p(n-1)}{(n-p)} F_{p,n-p}(\alpha)} = \sqrt{\frac{4 \times 14}{11} 3.36} = 4.14$$

versus

$$t_{n-1}(0.025) = 2.145.$$

Simultaneous versus one-at-a-time intervals

So the T^2 intervals are too wide when only applied to the p means, but the individual t-intervals are too narrow...



This leads us to considering the Bonferroni method.

Bonferroni Intervals

Assume we have m hypotheses to test or effects to estimate which can all be summarised through linear combinations of the form $\alpha_i' \mu$, $i = 1, \dots, m$.

Let C_i denote a confidence statement about the value of $\alpha_i' \mu$ with

$$\Pr[C_i \text{ true}] = 1 - \alpha_i, \quad i = 1, \dots, m.$$

Then the Bonferroni inequality leads to

$$\begin{aligned}\Pr[\text{all } C_i \text{ true}] &= 1 - \Pr[\text{at least one } C_i \text{ false}] \\ &\geq 1 - \sum_{i=1}^m \Pr(C_i \text{ false}) \\ &= 1 - \sum_{i=1}^m (1 - \Pr(C_i \text{ true})) \\ &= 1 - (\alpha_1 + \alpha_2 + \dots + \alpha_m)\end{aligned}$$

Bonferroni Intervals

Now let $\alpha_i = \frac{\alpha}{m}$ for the intervals $\bar{x} \pm t_{n-1}(\alpha_i/2)\sqrt{\frac{s_{ii}}{n}}$. Since

$$\Pr \left[\bar{X}_i \pm t_{n-1}(\alpha/2m) \sqrt{\frac{s_{ii}}{n}} \text{ contains } \mu_i \right] = 1 - \frac{\alpha}{m}$$

we have

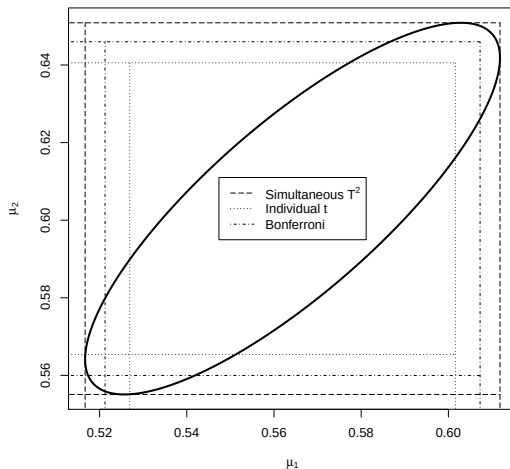
$$\Pr \left[\bar{X}_i \pm t_{n-1}(\alpha/2m) \sqrt{\frac{s_{ii}}{n}} \text{ contains all } \mu_i \right] \geq 1 - \alpha$$

So for p means, the Bonferroni confidence intervals will be of the form

$$\bar{x}_i - t_{n-1} \left(\frac{\alpha}{2p} \right) \sqrt{\frac{s_{ii}}{n}} \leq \mu_i \leq \bar{x}_i + t_{n-1} \left(\frac{\alpha}{2p} \right) \sqrt{\frac{s_{ii}}{n}}$$

Bonferroni Intervals

Example 5.4 (continued)



Large Sample Inferences about the Population Mean

Without the assumption of a normal population, when $n \gg p$ we base the confidence intervals on the χ^2 -distribution and so

$$\Pr \left[n(\bar{\mathbf{X}} - \boldsymbol{\mu})' \mathbf{S}^{-1} (\bar{\mathbf{X}} - \boldsymbol{\mu}) \leq \chi_p^2(\alpha) \right] \approx 1 - \alpha$$

and the intervals for linear combinations of the X variables are of the form

$$\mathbf{a}' \bar{\mathbf{X}} \pm \sqrt{\chi_p^2(\alpha)} \sqrt{\frac{\mathbf{a}' \mathbf{S} \mathbf{a}}{n}}$$

which leads to simultaneous confidence intervals for μ_i of the form

$$\bar{x}_i \pm \sqrt{\chi_p^2(\alpha)} \sqrt{\frac{s_{ii}}{n}}$$

Homework exercise 5.1 – Johnson & Wichern exercise 5.9

Measurements of $n = 61$ bears provided summary statistics (sample mean and sample covariance matrix) given in the file `J&WEx5.9.RData`. The variables, in order corresponding to the summary statistics, are:

Weight (kg) | Body length | Neck | Girth | Head length | Head width (all cm)

- a Obtain the large sample 95% simultaneous confidence intervals for the six population mean body measurements.
- b Obtain the large sample 95% simultaneous confidence ellipse for mean weight and mean girth.
- c Obtain the 95% Bonferroni confidence intervals for the six means in part a).
- d Refer to part b). Construct the 95% Bonferroni confidence rectangle for the mean weight and mean girth using $m = 6$. Compare this rectangle with the confidence ellipse in part b).
- e Obtain the 95% Bonferroni confidence interval for

mean head width – mean head length

using $m = 6 + 1 = 7$ to allow for this statement as well as statements about each individual mean.

Repeated Measures

Consider the following case:

- We take q repeated measures of some normally distributed variable (X) over time for n respondents.
- We want to test whether the μ' s at each time point are equal, or whether they change over time.

Arrange the data as $\mathbf{X} = [\mathbf{X}_1 \quad \cdots \quad \mathbf{X}_n]$, where each $\mathbf{X}_j = \begin{bmatrix} X_{1j} \\ X_{2j} \\ \vdots \\ X_{qj} \end{bmatrix}$ such that

X_{ij} is the i^{th} measurement on the j^{th} respondent.

Repeated Measures

We're interested in testing $H_0: \mu_1 = \mu_2 = \cdots = \mu_q$ vs the alternative that at least one μ_j differs.

This equivalent to testing

$$H_0: \mu_1 - \mu_2 = \mu_1 - \mu_3 = \cdots = \mu_1 - \mu_q = 0$$

or

$$H_0: \mu_1 - \mu_2 = \mu_2 - \mu_3 = \cdots = \mu_{q-1} - \mu_q = 0$$

Repeated Measures

We're interested in testing $H_0: \mu_1 = \mu_2 = \dots = \mu_q$ vs the alternative that at least one μ_j differs.

This equivalent to testing

$$H_0: \mu_1 - \mu_2 = \mu_1 - \mu_3 = \dots = \mu_1 - \mu_q = 0$$

or

$$H_0: \mu_1 - \mu_2 = \mu_2 - \mu_3 = \dots = \mu_{q-1} - \mu_q = 0$$

We can construct a contrast matrix containing contrast vectors $\mathbf{a}$ in the rows such that

$$\mathbf{C}_{(q-1) \times q} = \begin{bmatrix} 1 & -1 & 0 & \dots & 0 \\ 1 & 0 & -1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \dots & -1 \end{bmatrix} \text{ or } \mathbf{C}_{(q-1) \times q} = \begin{bmatrix} 1 & -1 & 0 & \dots & 0 \\ 0 & 1 & -1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & -1 \end{bmatrix}$$

Repeated Measures

The hypotheses above can now be expressed as

$$H_0: \mathbf{C}\boldsymbol{\mu} = \mathbf{0} \quad vs \quad H_1: \mathbf{C}\boldsymbol{\mu} \neq \mathbf{0}$$

for either of these contrast matrices $\mathbf{C}$.

The vector of differences is constructed as

$$\mathbf{Y} = \mathbf{C}\mathbf{X} \sim N_{q-1}(\mathbf{C}\boldsymbol{\mu}, \mathbf{C}\boldsymbol{\Sigma}\mathbf{C}')$$

such that the resulting Hotelling's T^2 statistic is

$$T^2 = n(\mathbf{C}\bar{\mathbf{X}})'(\mathbf{C}\mathbf{S}\mathbf{C}')^{-1}(\mathbf{C}\bar{\mathbf{X}}) \sim \frac{(n-1)(q-1)}{(n-q+1)} F_{q-1, n-q+1}$$

See the example in the notes (Example 6.10.1 on page 227 of Rencher) with the solution in `Lecture5.R`. Also attempt the following homework exercise.

Homework exercise 5.2 – Johnson & Wichern exercise 5.10

Refer to the bear growth data given in the file `T1-4.txt`, containing the weight (kg) and length (cm) of 7 female grizzly bears at ages 2, 3, 4 and 5 years respectively.

- a Obtain the 95% T^2 simultaneous confidence intervals for the four population means for length.
- b Refer to part a). Obtain the 95% T^2 simultaneous confidence intervals for the three successive yearly increases in mean length.
- c Obtain the 95% T^2 confidence ellipse for the mean increase in length from 2 to 3 years and the mean increase in length from 4 to 5 years.
- d Refer to parts a) and b). Construct the 95% Bonferroni confidence intervals for the set consisting of four mean lengths and three successive yearly increases in mean length.
- e Refer to parts c) and d). Compare the 95% Bonferroni confidence rectangle for the mean increase in length from 2 to 3 years and the mean increase in length from 4 to 5 years with the confidence ellipse produced by the T^2 -procedure.